# Bubble Chart

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## The Bubble chart

### *What is a bubble chart?*

A bubble chart is a scatter plot with an extra dimension. Like a scatter plot it places each data point at an (x, y) position, but it also varies the size of each point — the "bubble" — to encode a third value, and can use color for a fourth. It lets you see several variables at once in a single picture.

### *When to use it*

Use a bubble chart when you want to compare items on two measures and, at the same time, show how large or frequent each item is. For example: plot words by their length (x) against their average sentiment (y), with bubble size showing how often each word occurs — large bubbles sitting high or low jump out immediately.

### *The Bubble chart in the NLP Suite*

The bubble chart is available in the **Data visualization** GUI (select **Bubble chart** from the visualization dropdown). It reads a csv file and expects:

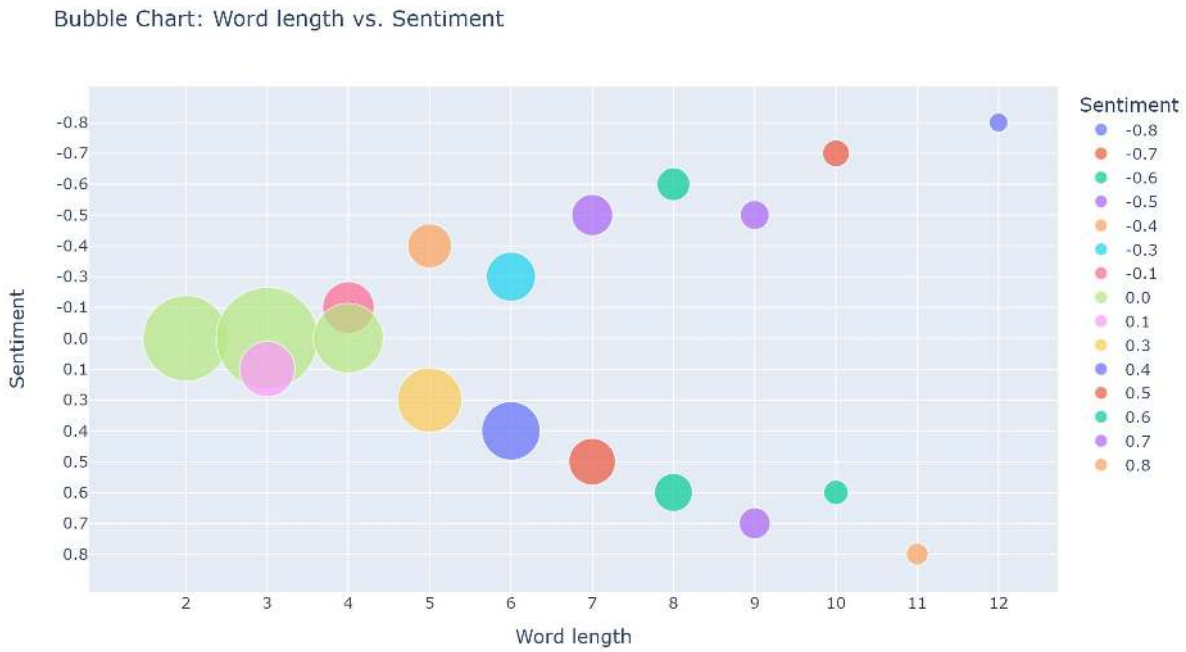
- **X-axis** — a csv field you select for the horizontal position.
- **Y-axis** — a csv field you select for the vertical position.
- **Bubble size** — the frequency of each value (how many times it occurs); larger counts make larger bubbles.
- **Color (optional)** — tick **Color** and enter an RGB value (e.g. 135, 207, 236) to set the bubble color.

**INPUT:** a csv file containing the fields you want on the two axes.

**OUTPUT:** an interactive Plotly HTML chart that opens in your browser. Hover over any bubble to read its exact values; use the toolbar (top-right) to zoom, pan, or save the chart as a PNG.

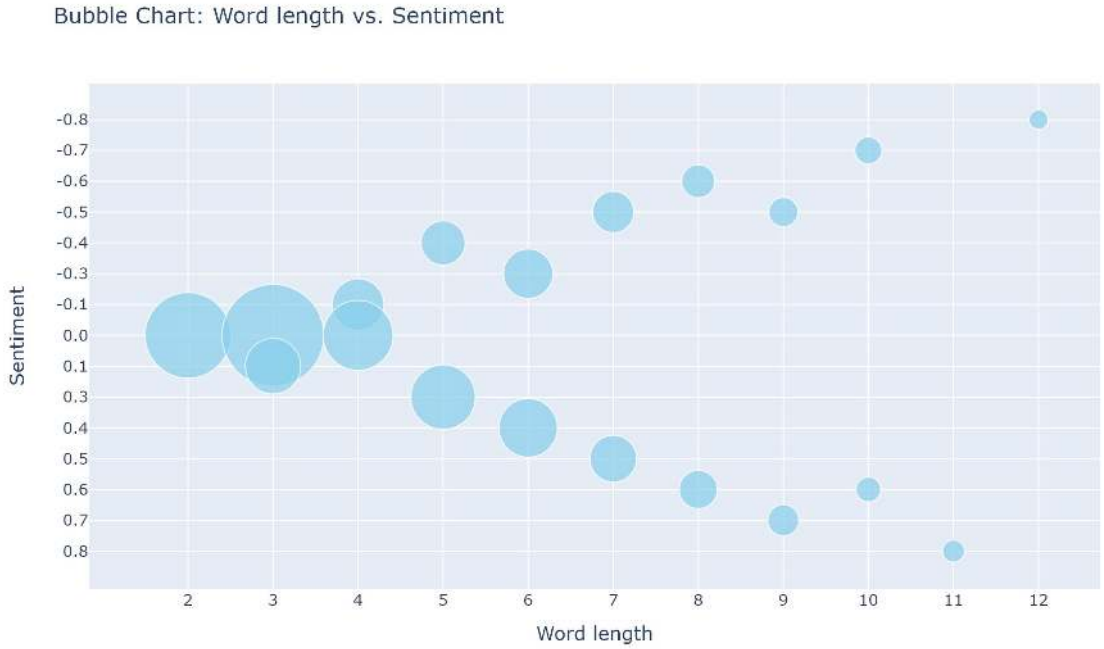
### *The bubble chart output*

The example below plots a corpus's words by length (x) against their rounded sentiment (y). Each bubble is one (length, sentiment) pair and its size is how many words share that pair, so the large neutral bubbles at short lengths (the frequent function words) stand out at a glance. By default the bubbles are automatically color-coded by the Y-axis value (a legend appears on the right).




*Bubble chart: word length × sentiment, bubble size = frequency (default color-coding by Y value).*

If you tick **Color** and enter an RGB value (e.g. **135, 207, 236**), every bubble is drawn in that single color instead — useful for a clean, uniform look or to match a house palette:




*The same chart with a single user-chosen bubble color (RGB 135, 207, 236).*

## *How to read it*

- **Position (x, y)** — the two measures you chose.
- **Size** — frequency: a bigger bubble means a more frequent value.
- **Color** — the single color you set, or the automatic color-coding by Y value.

Look for bubbles that are unusually large (very frequent) or that sit far from the crowd (outliers on your two measures).

### *Tip*

Because bubble size encodes frequency, a bubble chart is most useful when values repeat — word counts, category counts, and the like. If every row is unique (frequency 1), all bubbles are the same size and a plain scatter plot is clearer.

## Further readings — more TIPS files

TIPS_NLP_Charts - Sankey chart.pdf · TIPS_NLP_Charts - Treemap chart.pdf · TIPS_NLP_Statistical measures.pdf